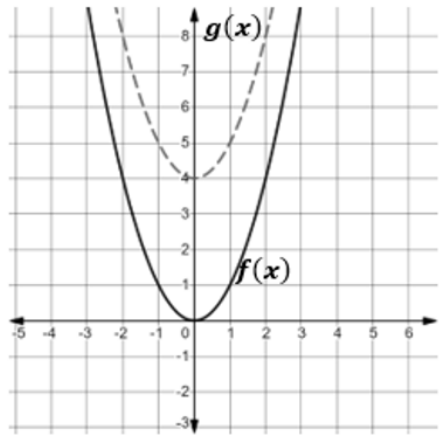
Algebra 1
Unit 13
Lesson 6 Practice

Name_Answer Key________
Date___________________
Period__________________

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



1. Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.

   Shift up 4 units

2. Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1	1	-1	5
0	0	0	4
1	1	1	5

3. Explain the change in the y-values from $f(x)$ to $g(x)$.

   y-values increase by 4.

4. Complete the statement:

   The graph of $g(x)$ is a [o horizontal / o vertical] shift of $f(x)$.

5. Complete the statement:

   The shift is seen in the table by [o adding / o subtracting] 4 to each [o x-value / o y-value] for each corresponding [o x-value. / o y-value.]

For 6-7, complete the statements to describe how the graph of $g(x) = f(x) - 5$ compares to the graph of $f(x) = |x + 2|$.

6. The graph of $g(x)$ is a

- o horizontal
- o vertical

shift of $f(x)$.

7. The graph of $g(x)$ is shifted __5__ units

- o up from
- o down from
- o to the right of
- o to the left of

the graph of $f(x)$.

8. Describe the effect on the x-values of $K(x) = |x - 3|$ for the transformation $J(x) = K(x) + 6$.

y-values will be shifted up for 6 for each corresponding x-value.

9. Describe the effect on the y-values of $v(x) = \frac{3}{4}x + 7$ for the transformation $w(x) = v(x) - 3$.

y-values will be shifted down 3 units.

10. Describe how the graph of $T(x) = S(x - 10)$ compares to the graph of $S(x) = x^2 + 2$.

$T(x)$ is shifted 10 units to the right.